1. Corporate Profile & Ontology Alignment: ApexBuild Construction Corp.

Document ID: DOC-01-CORP | **Version:** 4.2 (2026) | **Classification:** Official Corporate Standard
Primary OWL Classes: `ConstructionCompany`, `Customer`, `Employee`, `ManagementRole`, `Project`
Primary OWL Properties: `hasCustomer`, `hasEmployee`, `hasProject`, `hasSiteManager`, `hasDesigner`, `name`, `year`

1.1 Executive Overview of the Construction Company

ApexBuild Construction Corp. (represented in our formal domain ontology as an instance of class `ConstructionCompany`) is a Central European full-service civil engineering and general contracting enterprise founded in 1998 and headquartered in Bratislava, Slovakia, with regional operational hubs in Vienna and Prague. Under the formal axioms of the Construction Company Ontology, every valid `ConstructionCompany` must satisfy three existential restrictions simultaneously:

- `hasCustomer some Customer`: The company maintains active contractual relationships with verified private, corporate, and municipal clients (class `Customer`).
- `hasEmployee some Employee`: The company employs certified engineering, architectural, supervisory, and manual trade personnel (class `Employee`).
- `hasProject some Project`: The company executes at least one active or completed construction project (class `Project`).

Furthermore, in accordance with strict ontological disjointness rules, the class `ConstructionCompany` is mutually disjoint with `Person`, `Project`, `Material`, and `ConstructionMachine`. A legal corporate entity cannot simultaneously be classified as an individual person, a physical building project, a raw construction material, or a piece of heavy machinery.

1.2 Organizational Governance and Management Roles

Within ApexBuild Construction Corp., human resources are modeled under the top-level class `Person`, which carries the existential restriction `worksOn some Project`. Every employee of the firm belongs to the subclass `Employee` (which requires `hasProject some Project`). Specialized leadership and supervisory authority is formally grouped under the defined equivalent class `ManagementRole`:

> **Formal OWL Axiom (Equivalent Class):**
> `ManagementRole ≡ SiteManager ⊔ Architect` (The union of `SiteManager` and `Architect`).
> Only personnel classified as either a `SiteManager` or an `Architect` hold formal project sign-off authority, budget approval rights above EUR 250,000, and statutory responsibility on site.

Notice how object property hierarchy reflects organizational reporting lines: both `hasSiteManager` (range: `SiteManager`) and `hasDesigner` (range: `Designer`) are defined in the OWL ontology as sub-properties of `hasEmployee` (`rdfs:subPropertyOf hasEmployee`). Consequently, whenever a `ConstructionCompany` or `BuildingProject` is linked via `hasSiteManager` to an individual such as Ing. Martin Kováč, an OWL reasoner automatically infers that the individual is also linked via `hasEmployee`.

1.3 Key Corporate Divisions and Financial Metrics (2023–2026)

Division Name	Target Ontology Project Class	Division Lead (Role)	Annual Turnover (EUR)	Active Projects
Residential Development Division	`ResidentialProject`, `FamilyHouse`, `ApartmentBuilding`, `WoodenHouse`	Ing. Elena Varga (`Designer`) & Ing. Martin Kováč (`SiteManager`)	€48,500,000	14 projects
Commercial & High-Rise Division	`CommercialProject`, `OfficeBuilding`, `FamousProject`	Ing. arch. Richard Novák, PhD. (`Architect`)	€72,300,000	6 projects
Industrial & Logistics Division	`IndustrialProject`, `BuildingProject`	Ing. Peter Horváth (`SiteManager`)	€54,100,000	5 projects
Heritage & Reconstruction Division	`Reconstruction`, `HistoricalProject`, `MidCenturyProject`	Ing. Lukáš Beneš (`SiteManager`)	€19,800,000	8 projects

1.4 Strategic Client Portfolio (Class: Customer)

In the ontology, a `Customer` is a subclass of `Person` (representing either a natural person or a legal client representative) that satisfies the restriction `hasProject some Project` and `worksOn some Project` (participating in project reviews and handover). Additionally, every `BuildingProject` must satisfy `hasCustomer some Customer`, and every `Architect` satisfies `hasCustomer some Customer` because architects directly interface with commercial clients during feasibility and permitting phases.

Customer ID	Customer Name (`name`)	Client Segment	Contracted Projects (`hasProject`)	Assigned Architect / Site Manager
CUST-101	**J&T Real Estate Development Rep. (Mgr. Daniel Kráľ)**	High-Rise Residential	`EuroveaTower` (Year: 2023, Budget: €115M)	Ing. Martin Kováč (`SiteManager`)
CUST-102	**Danube Business Parks Rep. (Ing. Klára Šimková)**	Commercial Office	Danube Atrium `OfficeBuilding` (Year: 2024, Budget: €42M)	Ing. arch. Richard Novák (`Architect`)
CUST-103	**Dr. Tomáš Gregor & Family**	Private Residential	`FamilyHouseExample` in Záhorská Bystrica (Year: 2026)	Ing. Elena Varga (`Designer`)
CUST-104	**Alpine EcoHomes Rep. (Bc. Marek Zeman)**	Sustainable Timber Housing	Tatry CLT `WoodenHouse` Cluster (Year: 2025, Budget: €3.8M)	Ing. Elena Varga (`Designer`)
CUST-105	**Central European AutoLogistics Rep. (Ing. Robert Klein)**	Industrial Manufacturing	Trnava Smart Hall (`IndustrialProject`, Year: 2025)	Ing. Peter Horváth (`SiteManager`)

1.5 Classification of Projects by Chronological Period (ProjectPeriod)

ApexBuild Construction Corp. categorizes all historical, current, and planned engineering works using the ontology class `ProjectPeriod` and the datatype property `year` (`xsd:integer`). Using OWL `equivalentClass` definitions with numeric facet restrictions, any entity with a `year` value is automatically classified into one of four temporal categories:

- **`HistoricalProject`:** Equivalent to any project with `year < 1950` (`maxExclusive 1950`). Example: `EmpireStateBuilding` (`year = 1931`) studied as an architectural benchmark and heritage structural reference.
- **`MidCenturyProject`:** Equivalent to any project with `1950 ≤ year < 2000` (`minInclusive 1950` and `maxExclusive 2000`). Many of ApexBuild's current `Reconstruction` contracts target structures originally erected during this period.
- **`ModernProject`:** Equivalent to any project with `year ≥ 2000` (`minInclusive 2000`). Note that standard ontology classes `FamilyHouse`, `ApartmentBuilding`, `OfficeBuilding`, and `IndustrialProject` carry explicit restrictions requiring their completion/execution year to be at least 2000 (with standard catalog bounds `2000 ≤ year < 2026`, while specific 2026+ instances such as `FamilyHouseExample` extend into 2026).
- **`FutureProject`:** Equivalent to any project with `year ≥ 2027` (`minInclusive 2027`). This covers next-generation net-zero industrial complexes and hydrogen-ready commercial towers scheduled in the 2027–2030 pipeline.